

1 Analysis of the given model

The given model was trained using the following hyperparameters: $epochs = 100$, $batch\ size = 128$, $learning\ rate = 0.001$ and $optimizer = Adam$. Loss figure mainly shows overfitting. This problem could be solved by introducing regularization layers and potentially adjusting the learning rate.

2 Optimization

Due to the large size of the network and the number of features in the dataset, the first step to reduce overfitting was to apply a regularization technique such as L1 or L2. This choice was made after analyzing the model's weights. Most of the layers have very small weights, with an average close to zero. Therefore, using L1 regularization, which performs feature selection and creates a sparse weight distribution, would not be useful. If applied, it would make the weights even smaller and prevent the network from learning properly. For this reason, L2 and L1L2 regularization, with $\alpha = 0.01$ were chosen.

L1L2 regularization was applied to layers two, five, six, and seven. This choice was made for two reasons: the first is that layers two and five are conceptually redundant with (or similar to) layers one and four. This structural similarity can create redundancy in the learned features. The second reason is that layers five and six have some weights that are quite large, but applying only L1 regularization would risk forcing all the layer weights to zero, which would prevent learning. Therefore, the combined L1L2 penalty was used to encourage sparsity (L1 feature), while also penalizing large weights to exist (L2 feature).

In layers one, three and four the weight distribution was already sparse. However, since there were some large weights L2 regularization was applied in these layers to limit them and obtain better performance. Moreover, Dropout layers with a rate of 0.2 were added throughout the network because the model is very dense and this helps to simplify the learning process and reduce overfitting.

The obtained results were not good. By plotting the model weights it is clear that L1L2 and L2 regularizations were too strong. The regularization constant α was reduced, to $\alpha = 0.00001$ for L1L2 and $\alpha = 0.0001$ for L2. Observing the loss figure of *first model improved* it was also clear that the learning rate was too high, it was decreased to 0.0001.

The results of the *second improved model* were quite good, the loss curve showed overfitting and suggested a slightly low learning rate. A new model was created using a learning rate of 0.0005. The loss curve of the *third improved model* did not show overfitting but in the accuracy and loss curves there is a clear situation of underfitting due to: the low accuracy value and the high loss value (~ 1.6). Analysis of the weights showed a sparse distribution in layer one, three, four and five.

At this point, it was concluded that having sparse weights might not be a problem. Probably the architecture of the model is not enough to obtain better performance and other architectures like convolutional neural networks would be better to solve this problem. Nevertheless, one final attempt was made to get better performance by applying L1L2 regularization instead of L2 regularization in layers one, three, four, and five. The obtained results of the *fourth improved model* were worse, and the loss curve again suggested a slightly low learning rate. A *fifth improved model* was created, but there was no improvement; the model was still in a situation of underfitting.

It is likely that the sparse weight distribution of the *third improved model* was working as a form of feature selection, and that by applying L1L2 regularization, performance dropped (accuracy fell from 50% to 40%). The best model to solve this classification problem was the *third improved model*, even if its performance was not great for the reasons discussed before.